

CYCOLOY* CX2244ME Resin

Monday, October 31, 2011

General Information

Product Description

Cycloy* CX2244ME resin is a flame retardant PC/ABS blend featuring excellent balance of flow and impact together with UL-94 V0 rating at < 1mm. Cycloy CX2244ME resin also demonstrates improved chemical resistance to general hospital cleaners which can be considered for healthcare enclosure applications.

General

Material Status	• Commercial: Active	
Availability	• North America	
Additive	• Flame Retardant	
Features	• Flame Retardant • Good Chemical Resistance	• Good Flow • Good Impact Resistance
Uses	• Hospital Goods	• Housings
Processing Method	• Injection Molding	

ASTM & ISO Properties ¹

Physical	Nominal Value	Unit	Test Method
Specific Gravity	1.19	g/cm ³	ASTM D792
Density	1.20	g/cm ³	ISO 1183
Melt Mass-Flow Rate (MFR) (260°C/2.16 kg)	18	g/10 min	ASTM D1238
Melt Volume-Flow Rate (MVR) (260°C/2.16 kg)	15.0	cm ³ /10min	ISO 1133
Molding Shrinkage - Flow (3.20 mm)	0.40 to 0.60	%	Internal Method
Water Absorption (Saturation, 23°C)	0.20	%	ISO 62
Water Absorption (Equilibrium, 23°C, 50% RH)	0.10	%	ISO 62
Mechanical	Nominal Value	Unit	Test Method
Tensile Modulus ²	2600	MPa	ASTM D638
Tensile Modulus	2600	MPa	ISO 527-2/1
Tensile Strength ³ (Yield)	65.0	MPa	ASTM D638
Tensile Stress (Yield)	65.0	MPa	ISO 527-2/50
Tensile Strength ³ (Break)	58.0	MPa	ASTM D638
Tensile Stress (Break)	50.0	MPa	ISO 527-2/50
Tensile Elongation ³ (Yield)	4.1	%	ASTM D638
Tensile Strain (Yield)	4.0	%	ISO 527-2/50
Tensile Elongation ³ (Break)	100	%	ASTM D638
Tensile Strain (Break)	90	%	ISO 527-2/50
Flexural Modulus ⁴ (50.0 mm Span)	2500	MPa	ASTM D790
Flexural Modulus ⁵	2500	MPa	ISO 178
Flexural Strength ^{5,6}	96.0	MPa	ISO 178
Flexural Strength ⁴ (Yield, 50.0 mm Span)	104	MPa	ASTM D790

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Impact	Nominal Value	Unit	Test Method
Charpy Notched Impact Strength ⁷			ISO 179/1eA
-30°C	10	kJ/m ²	
23°C	40	kJ/m ²	
Notched Izod Impact			ASTM D256
-30°C	180	J/m	
23°C	700	J/m	
Notched Izod Impact Strength ⁸			ISO 180/1A
-30°C	10	kJ/m ²	
23°C	40	kJ/m ²	
Instrumented Dart Impact (23°C, Total Energy)	65.0	J	ASTM D3763
Thermal	Nominal Value	Unit	Test Method
Deflection Temperature Under Load			ASTM D648
1.8 MPa, Unannealed, 3.20 mm	89.0	°C	
Heat Deflection Temperature ⁹			ISO 75-2/Af
1.8 MPa, Unannealed, 64.0 mm Span	93.0	°C	
Vicat Softening Temperature	110	°C	ASTM D1525 ¹⁰
Vicat Softening Temperature			
--	110	°C	ISO 306/B50
--	113	°C	ISO 306/B120
Ball Pressure Test (75°C)	Pass		IEC 60695-10-2
CLTE - Flow (-40 to 40°C)	0.000075	cm/cm/°C	ASTM E831
CLTE - Flow (-40 to 40°C)	0.000075	cm/cm/°C	ISO 11359-2
CLTE - Transverse (-40 to 40°C)	0.000075	cm/cm/°C	ASTM E831
CLTE - Transverse (-40 to 40°C)	0.000075	cm/cm/°C	ISO 11359-2
Flammability	Nominal Value	Unit	Test Method
Flame Rating - UL			UL 94
0.750 mm	V-0		
1.50 mm	5VB		
Glow Wire Flammability Index (3.00 mm)	960	°C	IEC 60695-2-12
Glow Wire Ignition Temperature (3.00 mm)	800	°C	IEC 60695-2-13
UL	Nominal Value	Unit	Test Method
RTI Str	90.0	°C	UL 746
RTI Imp	90.0	°C	UL 746
RTI Elec	90.0	°C	UL 746
Comparative Tracking Index (CTI) (PLC)	PLC 3		UL 746
Hot-wire Ignition (HWI) (PLC)	PLC 3		UL 746
High Amp Arc Ignition (HAI) (PLC)	PLC 0		UL 746

Processing Information

Injection	Nominal Value	Unit
Drying Temperature	80.0 to 90.0	°C
Drying Time	2.0 to 4.0	hr

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Injection	Nominal Value	Unit
Suggested Max Moisture	0.020	%
Hopper Temperature	60.0 to 80.0	°C
Rear Temperature	230 to 280	°C
Middle Temperature	240 to 290	°C
Front Temperature	250 to 300	°C
Nozzle Temperature	250 to 300	°C
Processing (Melt) Temp	250 to 300	°C
Mold Temperature	60.0 to 85.0	°C
Vent Depth	0.030 to 0.075	mm

Notes

¹ Typical properties: these are not to be construed as specifications.

² 5.0 mm/min

³ Type I, 50 mm/min

⁴ 1.3 mm/min

⁵ 2.0 mm/min

⁶ Yield

⁷ 80*10*3 sp=62mm

⁸ 80*10*3

⁹ 80*10*4 mm

¹⁰ Rate B (120°C/h), Loading 2 (50 N)

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